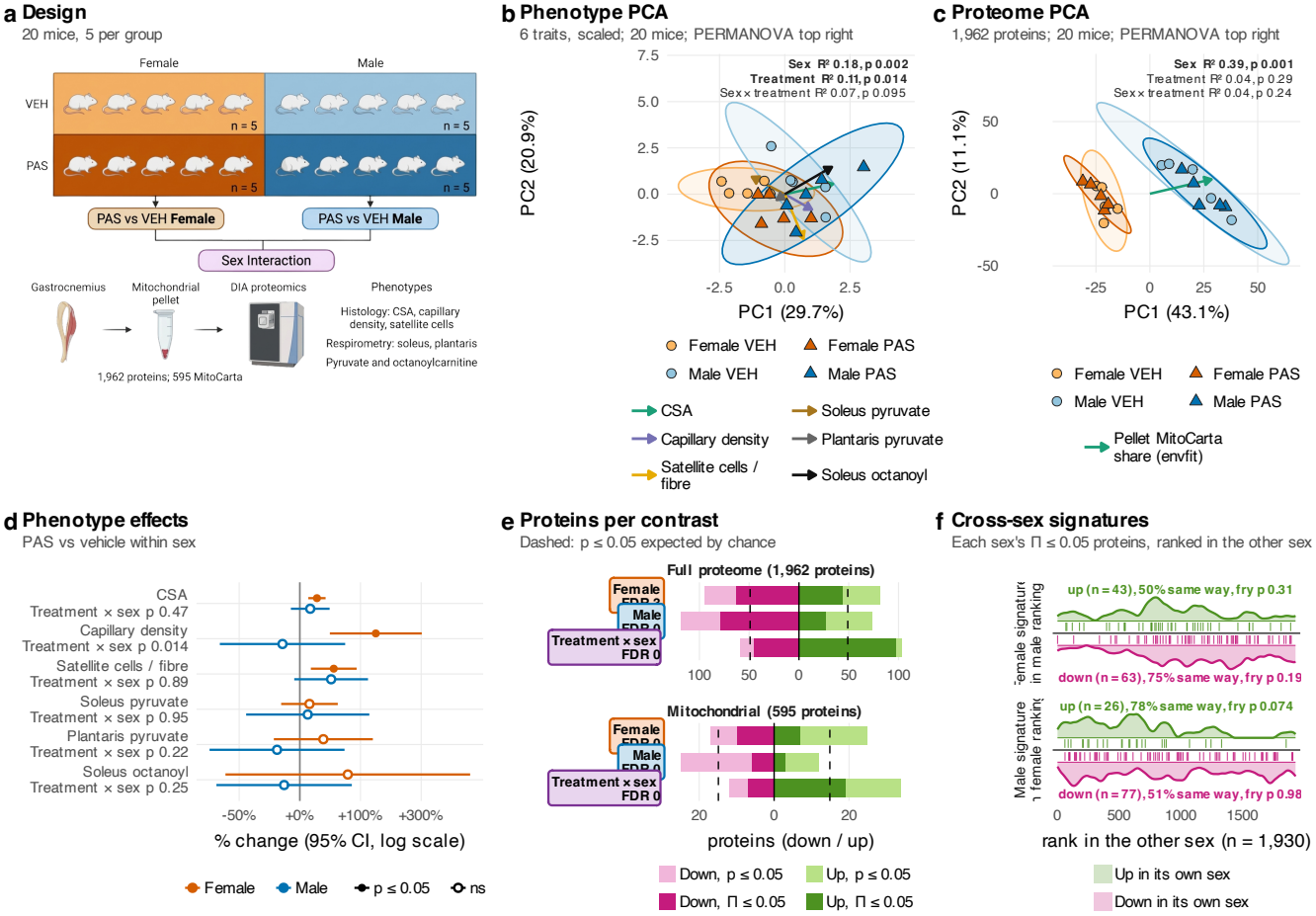

a Design
20 mice, 5 per group
Female
Male
VEH
PAS
n = 5
PAS vs VEH Female
PAS vs VEH Male
Sex Interaction
Gastrocnemius
Mitochondrial pellet
DIA proteomics
1,962 proteins; 595 MitoCarta
Phenotypes
Histology: CSA, capillary density, satellite cells
Respirometry: soleus, plantaris
Pyruvate and octanoylcarnitine
b Phenotype PCA
6 traits, scaled; 20 mice; PERMANOVA top right
Sex R² 0.18, p 0.002
Treatment R² 0.11, p 0.014
Sex × treatment R² 0.07, p 0.095
PC2 (20.9%)
PC1 (29.7%)
Female VEH
Female PAS
Male VEH
Male PAS
CSA
Soleus pyruvate
Capillary density
Plantaris pyruvate
Satellite cells / fibre
Soleus octanoyl
c Proteome PCA
1,962 proteins; 20 mice; PERMANOVA top right
Sex R² 0.39, p 0.001
Treatment R² 0.04, p 0.29
Sex × treatment R² 0.04, p 0.24
PC2 (11.1%)
PC1 (43.1%)
Female VEH
Female PAS
Male VEH
Male PAS
Pellet MitoCarta share (envfit)
d Phenotype effects
PAS vs vehicle within sex
CSA
Treatment × sex p 0.47
Capillary density
Treatment × sex p 0.014
Satellite cells / fibre
Treatment × sex p 0.89
Soleus pyruvate
Treatment × sex p 0.95
Plantaris pyruvate
Treatment × sex p 0.22
Soleus octanoyl
Treatment × sex p 0.25
-50%
+0%
+100%
+300%
% change (95% CI, log scale)
Female
Male
p ≤ 0.05
ns
e Proteins per contrast
Dashed: p ≤ 0.05 expected by chance
Full proteome (1,962 proteins)
Female FDR 0
Male FDR 0
Treatment × sex FDR 0
Mitochondrial (595 proteins)
proteins (down / up)
Down, p ≤ 0.05
Up, p ≤ 0.05
Down, Π ≤ 0.05
Up, Π ≤ 0.05
f Cross-sex signatures
Each sex's Π ≤ 0.05 proteins, ranked in the other sex
up (n = 43), 50% same way, fry p 0.31
down (n = 63), 75% same way, fry p 0.19
up (n = 26), 78% same way, fry p 0.074
down (n = 77), 51% same way, fry p 0.98
Female signature in male ranking
Male signature in female ranking
rank in the other sex (n = 1,930)
Up in its own sex
Down in its own sex